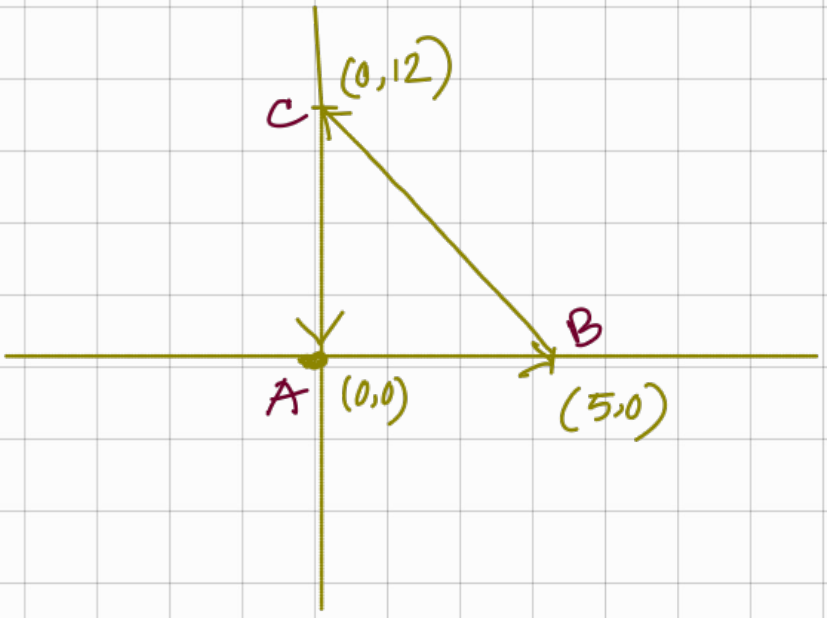
① $\begin{bmatrix} c \\ d \end{bmatrix} = m \begin{bmatrix} a \\ b \end{bmatrix}$ → Linearly dep. rank 1.

$c = ma$, $d = mb$

$\therefore m = \frac{c}{a}$ $\therefore d = \frac{c}{a} b$ $\therefore ad = cb$ (condⁿ)(ans)

②



$u = B - A = (5,0) - (0,0) = (5,0)$

$v = B - C = (0,12) - (5,0) = (-5,12)$

$w = A - C = (0,0) - (0,12) = (0,-12)$

x components $= 5 - 5 + 0 = 0$

y " $= 0 + 12 - 12 = 0$

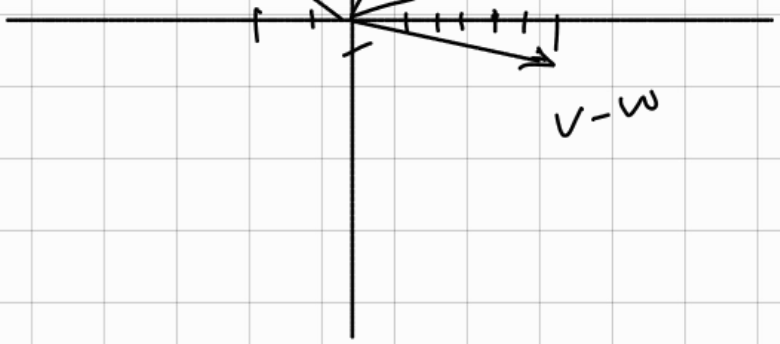
③ Line, Plane in R^3, all of R^3

④

$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$w = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$





$v+w=\begin{bmatrix}2\\3\end{bmatrix}$

$v-w=\begin{bmatrix}6\\-1\end{bmatrix}$

⑤ $v+w=\begin{bmatrix}5\\1\end{bmatrix}$ $\quad v-w\begin{bmatrix}1\\5\end{bmatrix}$

$v+w-v-w=\begin{bmatrix}6\\-6\end{bmatrix}$

$2v=\begin{bmatrix}6\\6\end{bmatrix} \quad \therefore v=\begin{bmatrix}3\\3\end{bmatrix}$

$w=\begin{bmatrix}5\\1\end{bmatrix}-\begin{bmatrix}3\\3\end{bmatrix}=\begin{bmatrix}2\\-2\end{bmatrix}$

⑥ $v=\begin{bmatrix}2\\1\end{bmatrix}, \; w=\begin{bmatrix}1\\2\end{bmatrix} \quad \therefore cv+dw=2c+d, c+2d$

$c\begin{bmatrix}2\\1\end{bmatrix}+d\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}2c+d\\c+2d\end{bmatrix}$

⑦ $c\begin{bmatrix}1\\2\\3\end{bmatrix}+d\begin{bmatrix}-3\\1\\-2\end{bmatrix}=\begin{bmatrix}2\\-3\\-1\end{bmatrix}$ ($\vec{u}$, $\vec{v}$, $\vec{w}$)

$(c-3d=2)\times 2$

$2c+d=-3$

$3c-2d=-1$

$7c+7d=-7$

$d=-1$

closed loop (polygon)

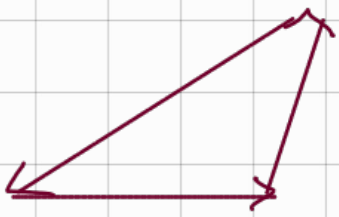
$u+v+w=0$

↓

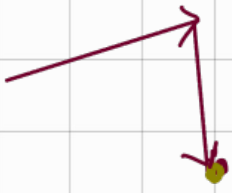
So same plane.

$c = -1$

$u = -v - u$ is in the plane of v & u



no way to form on a 3d. always 2D, use intuition.



pointing to screen (not on the same plane)

(8) $v = (1, -2, 1)$, $w = (0, 1, -1)$

$$cv + dw = c(1, -2, 1) + d(0, 1, -1)$$

$c \cdot 1 + 0$
$-2c + d$
$c - d$

sum components of $v = 1 - 2 + 1 = 0$
" " " $w = 0 + 1 - 1 = 0$

$$c - 2c + c + d - d = 0$$

So, every vector written as $cv + dw$ must have compo that sum to zero.

$(3, 3, -6)$ check: $c = 3$, $d = 9$

$$3v + 9w = (3, -6, 3) + (0, 9, -9)$$

$= 3, 3, 6$

$(3,3,6) \not\in \rightarrow$ 12 not 0

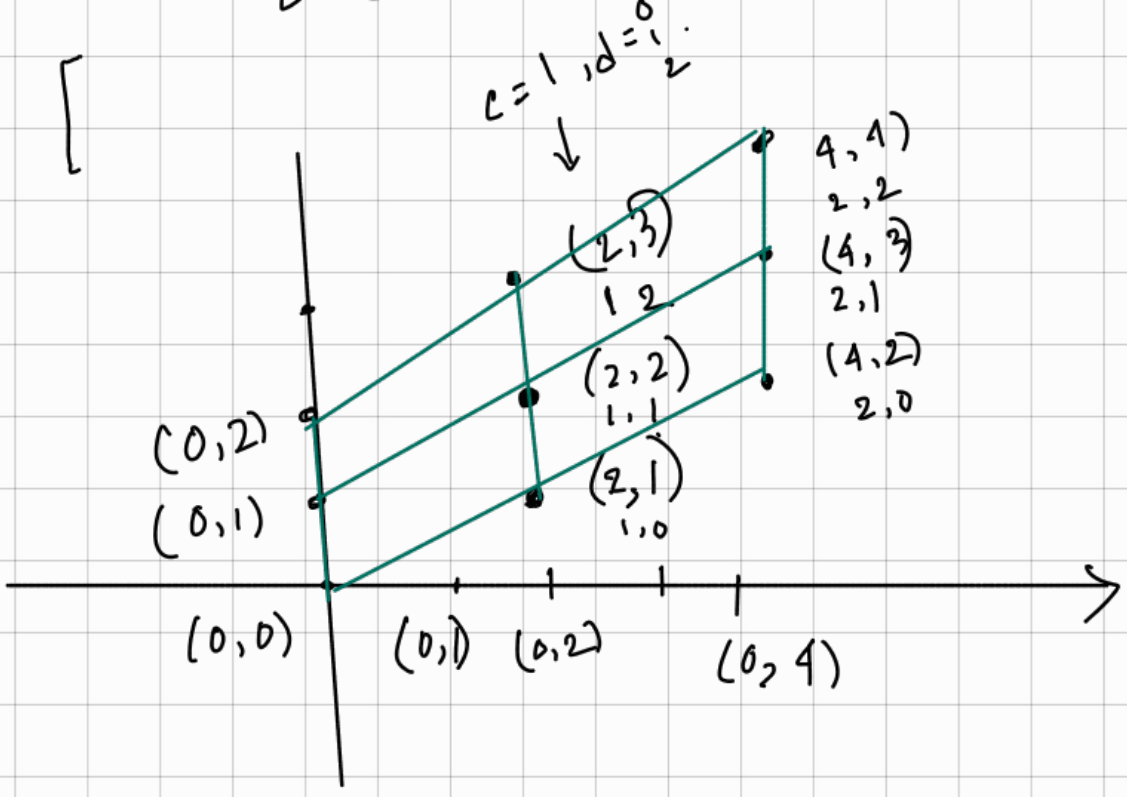
so no soln to $cv + dw = (3,3,6)$

⑨ $0\begin{bmatrix}2\\1\end{bmatrix} + 0\begin{bmatrix}0\\1\end{bmatrix} = 0$ $\quad c = 0, 1, 2$ $\quad d = 0, 1, 2$

$0\begin{bmatrix}2\\1\end{bmatrix} + 1\begin{bmatrix}0\\1\end{bmatrix} = 0, d$

$2\begin{bmatrix}0\\1\end{bmatrix} = 0, 2d$

$1\,[$



$$c\begin{bmatrix}2\\1\end{bmatrix} + d\begin{bmatrix}0\\1\end{bmatrix} = 0 = \begin{bmatrix}2c\\c+d\end{bmatrix}$$

⑩ decide $u = (1,1,0)$, $v = (0,1,1)$ & $w = (a,b,c)$ linearly dep or indep.

$\Rightarrow$ if (a,b,c) is a combination of

$cu + dv$

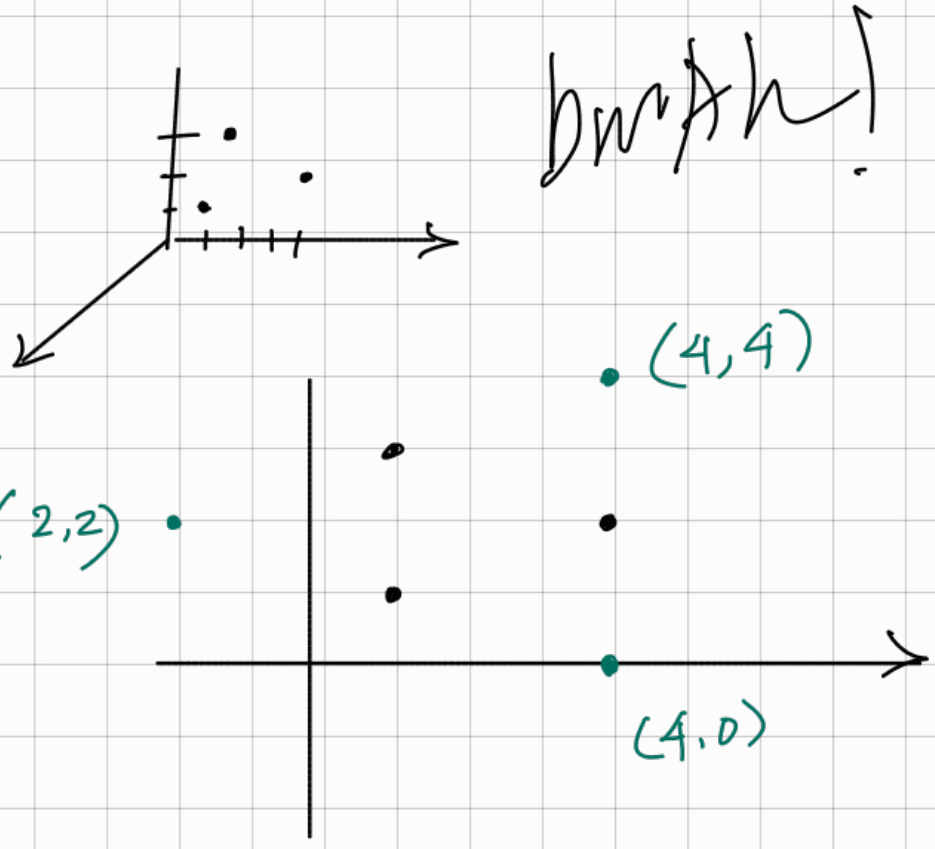
$$c\begin{bmatrix}1\\1\\0\end{bmatrix} + d\begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}a\\b\\c\end{bmatrix}$$

$c = a$, $d = c$

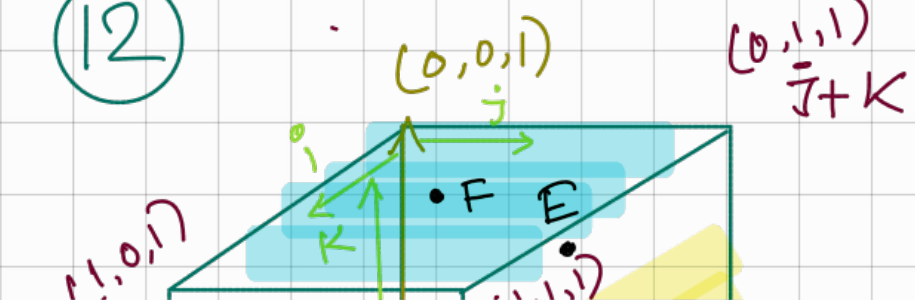
so, middle component gives requirement:

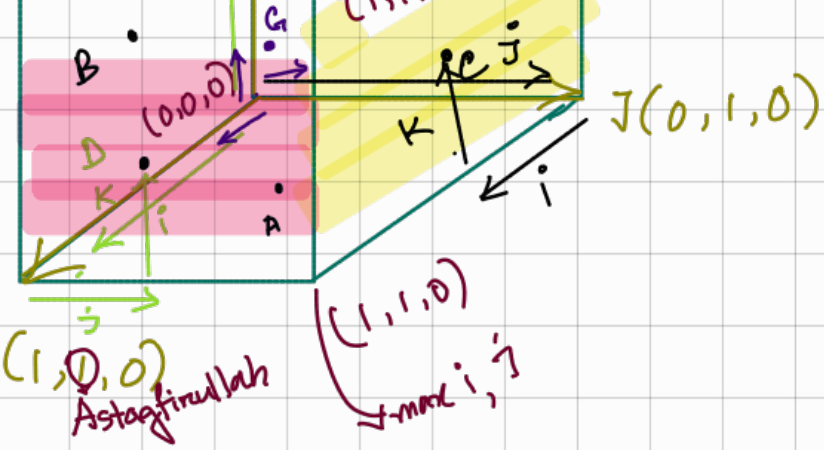
$$\boxed{a + c = b}$$

(11)



(12)



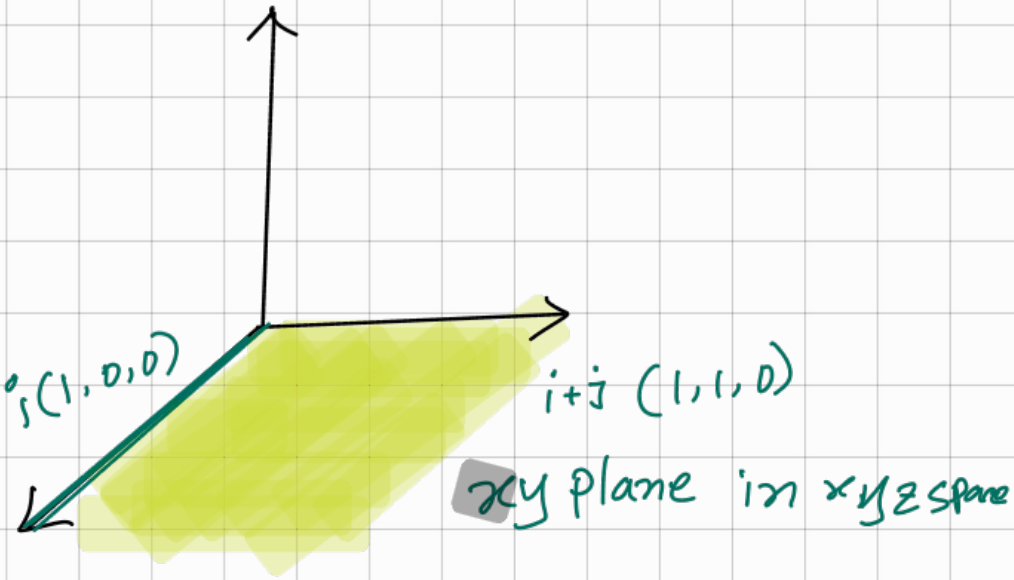


$A = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$ $B = \left(\frac{1}{2}, 0, \frac{1}{2}\right)$ $C = \left(\frac{1}{2}, 1, \frac{1}{2}\right)$

$D = \left(1, \frac{1}{2}, \frac{1}{2}\right)$ $E = \left(0, \frac{1}{2}, \frac{1}{2}\right)$, $F = \left(\frac{1}{2}, \frac{1}{2}, 1\right)$

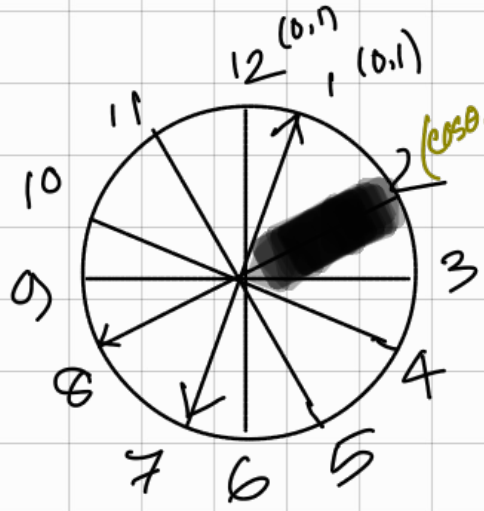
$G = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

(13)



xy plane in xyz space

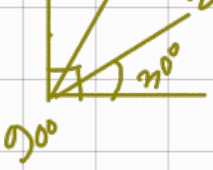
(14)



a) sum of vect 1:00, 2:00, ... 12:00 = 0.

b) 2 removed, all other pairs cancel out. So 8:00 single remains.

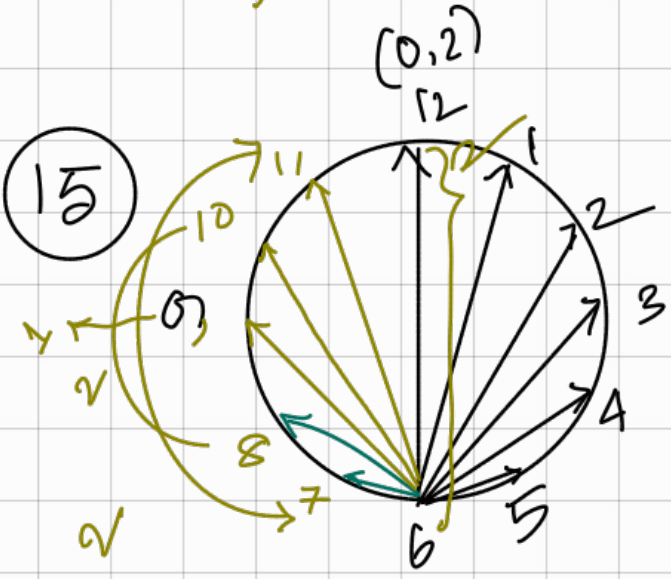
(c) 30°.



$\cos\theta = \cos 30°, \sin 30°$

$= \frac{\sqrt{3}}{2}, \frac{1}{2}$

(15)



moving origin to 6:00 adds $\hat{j} = (0,1)$ to every vector

So, sum $= 12\hat{j} = (0,12)$

nothing cancels y.



$2+2+1+2+2+1$

$= 10 + 2 + 0$

(16) $cu + dv + ew \longrightarrow$ line

$cu + dcu + ecu \longrightarrow$ all multiple of u.

→ plane

$$\boxed{cu + dv + ew} = 0$$

(17)

$$c\begin{bmatrix}1\\2\end{bmatrix} + d\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}14\\8\end{bmatrix}$$

$c + 3d = 14$

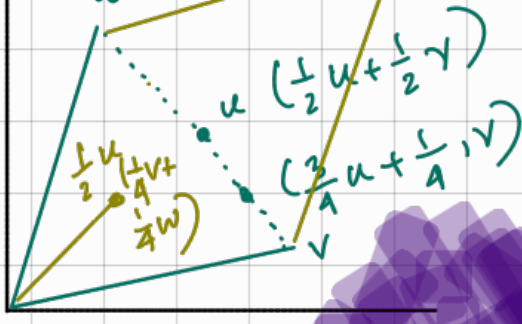
$2c + d = 8$

$0c - 5d = 20 \quad \therefore d = 4$

$c = 2$

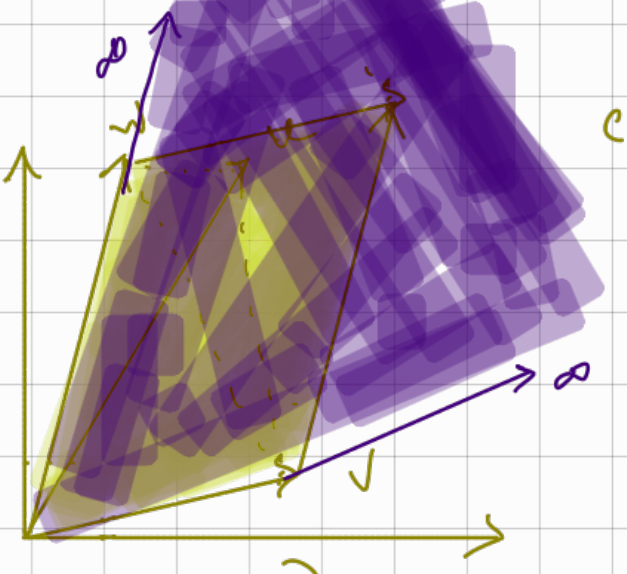
$2u$ $(v+w)$

(18)



the cone.

(19)



$cv+dw$ → $1v$ at most

can't be greater than 1

$0 \le c \le 1$ → $1w$ "

$0 \le d \le 1$

$c \ge 0$? } cone

$d \ge 0$ } ↓

whole 1st quad

$V=(a,0)$, $W(b,0)$

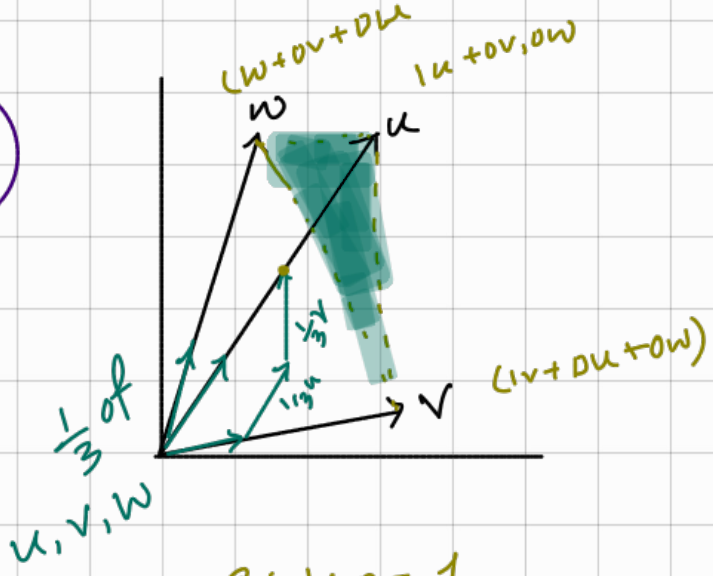
→ segment of a line

$W=-V$ → cone opens to a half space

$V(1,0)$, $W(-1,0)$ → a line

fill

(20)



$\frac{1}{3}$ of

u, v, w

$c+d+e=1$

return for challenge probs.